Fake News Detection Using Transformer Models and Explainable AI

Hemarathan Munnangi
Dept. of Computer Engineering
California State University, Long Beach
Long Beach, CA, USA
hemarathan.munnangi01@student.csulb.edu

Mauli Arunkumar Pandya
Dept. of Computer Engineering
California State University, Long Beach
Long Beach, CA, USA
MauliArunkumar.Pandya01@student.csulb.edu

Mansi Bharatbhai Lathiya
Dept. of Computer Engineering
California State University, Long Beach
Long Beach, CA, USA
mansibharatbhai.lathiya01@student.csulb.edu

Gyanendra Pandey
Dept. of Computer Engineering
California State University, Long Beach
Long Beach, CA, USA
gyanendra.pandey01@student.csulb.edu

***Abstract*—The rapid proliferation of misinformation across online platforms poses a serious threat to public discourse and democratic processes. This paper presents a fake news detection system that combines transformer-based language models with Explainable AI (XAI) techniques to achieve both high accuracy and interpretability. We fine-tune a pre-trained DistilBERT model on the Kaggle Fake-and-Real-News dataset comprising 44,034 cleaned articles, and compare its performance against two traditional baselines: TF-IDF with Logistic Regression and TF-IDF with Linear SVM. The fine-tuned DistilBERT model achieves an F1-score of 0.9995 and perfect precision of 1.0000 on the test set, outperforming both baselines. However, these results may be partially influenced by dataset-specific artifacts such as source attribution patterns, which may limit generalization to other domains. To address the black-box limitation of transformer models, we apply both LIME and SHAP post-hoc explanation methods to identify the words and tokens most influential in driving model predictions. Our results demonstrate that DistilBERT captures rich contextual patterns that surface-level TF-IDF features cannot, while the XAI component provides meaningful transparency into model decision-making. The complete codebase is publicly available on GitHub.**

***Index Terms*—fake news detection, natural language processing, DistilBERT, transformer models, explainable AI, LIME, SHAP, misinformation**

I. Introduction

The widespread dissemination of fake news through social media and online news platforms has emerged as one of the most pressing challenges of the digital age. Misinformation influences public opinion, undermines trust in credible journalism, and can have serious real-world consequences ranging from public health crises to political instability [1]. Because fabricated articles often mimic the style and vocabulary of legitimate journalism, manual fact-checking at scale is impractical.

Automated fake news detection using Natural Language Processing (NLP) has therefore attracted significant research attention. Early approaches relied on handcrafted lexical features combined with traditional classifiers such as Logistic Regression and Support Vector Machines [2]. While computationally efficient, these methods fail to capture the deeper semantic and contextual patterns that distinguish real news from fabricated content, particularly as fake news writing styles evolve.

The advent of transformer-based pre-trained language models such as BERT [8] has dramatically advanced NLP performance across a wide range of tasks. Fine-tuned transformer models have been shown to consistently outperform traditional and recurrent approaches for fake news detection [11]. However, these models operate as black boxes, offering little transparency into their decision-making process. This opacity reduces trust among end users such as journalists, fact-checkers, and platform moderators.

This paper addresses these challenges by proposing a fake news detection pipeline that combines the contextual language understanding of DistilBERT [13] with post-hoc Explainable AI techniques — specifically LIME [14] and SHAP [15] — to produce both accurate and interpretable predictions. Our system is evaluated on the Kaggle Fake-and-Real-News dataset and compared against two strong TF-IDF baselines.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the dataset and proposed methods. Section IV presents experimental results. Section V discusses findings and limitations. Section VI concludes the paper with directions for future work.

II. Literature Review

A. Traditional Machine Learning Approaches

Early fake news detection research relied on handcrafted textual features such as bag-of-words and TF-IDF representations combined with classifiers including Logistic Regression, Naïve Bayes, and Support Vector Machines [2]. The introduction of the LIAR dataset by Wang [4] provided a widely used benchmark for evaluating such approaches. Although computationally efficient, these models rely on surface-level lexical features and struggle to capture deeper semantic relationships, limiting their generalization across evolving writing styles [5].

B. Recurrent Neural Networks

To capture sequential context, researchers applied Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks to fake news detection. Rashkin et al. [6] demonstrated that neural language models capture subtle linguistic differences between propaganda, satire, and legitimate news. Mallik and Kumar [7] combined Word2Vec embeddings with LSTM for context-aware detection. However, RNNs struggle with long-range dependencies, motivating the adoption of attention-based transformer architectures.

C. Transformer-Based Models

Devlin et al. [8] introduced BERT, which uses bidirectional self-attention to capture rich contextual word representations. BERT-based models have since been widely applied to fake news detection. Kaliyar et al. [9] proposed FakeBERT, combining BERT embeddings with convolutional layers. Farokhian et al. [10] proposed a dual-BERT architecture processing headlines and article bodies separately. Raza et al. [11] demonstrated that fine-tuned transformer models consistently outperform earlier approaches. DistilBERT [13] retains 97% of BERT's performance while being 40% smaller and 60% faster, making it well suited for practical deployment.

D. Large Language Models

The emergence of large language models (LLMs) such as GPT-based systems has introduced new challenges, as LLM-generated misinformation can be difficult for traditional detectors to identify [3]. While LLMs have been explored as classifiers [12], their computational cost and reliance on proprietary APIs limit practical deployment. Lightweight transformer models such as DistilBERT remain more accessible alternatives.

E. Explainable AI for Fake News Detection

Despite strong predictive performance, transformer-based models lack interpretability. LIME [14] and SHAP [15] are widely used post-hoc explanation methods that identify features most influential in model predictions. Albtoush et al. [16] proposed an explainability framework for BERT-based detectors, while Hashmi et al. [17] demonstrated that XAI methods improve human trust in automated detection systems. However, integrating explainability with efficient transformer classifiers remains an active research challenge.

F. Research Gaps

Three key gaps motivate this work. First, many high-performing transformer-based detectors do not provide interpretable explanations. Second, studies often prioritize accuracy over transparency, which is critical for user trust in misinformation detection. Third, computationally expensive models such as LLMs are difficult to deploy practically. This work addresses all three gaps by combining DistilBERT with both LIME and SHAP explainability within a reproducible pipeline.

III. Methods

A. Dataset

We use the Kaggle Fake-and-Real-News dataset [18], which contains news articles collected from two sources: fake articles from unreliable websites flagged by PolitiFact and real articles from Reuters. The dataset consists of 23,481 fake articles (label 0) and 21,417 real articles (label 1), totaling 44,898 articles across five columns: title, text, subject, date, and label. Fig. 1 shows the class distribution, confirming a near-balanced dataset requiring no class balancing techniques.

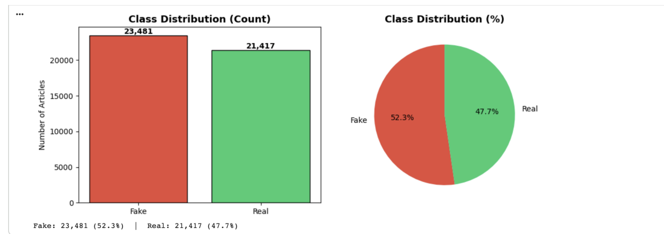

Fig. 1. Class distribution of the dataset: 23,481 Fake (52.3%) and 21,417 Real (47.7%).

B. Preprocessing

Raw text was preprocessed through the following steps: (1) duplicate removal, reducing the dataset from 44,898 to 44,689 articles; (2) lowercasing and whitespace normalization; (3) removal of special characters while retaining basic punctuation; (4) truncation of each article to the first 500 words to manage input length; and (5) removal of articles with fewer than 10 characters after cleaning. The final preprocessed dataset contains 44,034 articles with a near-balanced class distribution of 51.8% fake and 48.2% real.

C. Data Splitting

The dataset was partitioned into training (80%), validation (10%), and test (10%) sets using stratified random sampling with a fixed random seed of 42 to ensure reproducibility and preserve class distribution across all splits. This yielded 35,227 training articles, 4,403 validation articles, and 4,404 test articles.

D. Baseline Models

Two TF-IDF-based baseline models were implemented using scikit-learn pipelines. The TF-IDF vectorizer was configured with a vocabulary limit of 50,000 features, unigram and bigram representations (ngram_range = (1,2)), sublinear term frequency scaling, and a minimum document frequency of 2. Two classifiers were evaluated: Logistic Regression with regularization parameter C = 1.0 and maximum iterations of 1,000, and Linear SVM with C = 1.0 and maximum iterations of 2,000.

E. DistilBERT Fine-Tuning

The main model is based on the `distilbert-base-uncased` checkpoint from HuggingFace [13], which comprises 66,955,010 trainable parameters organized across 6 transformer layers with bidirectional self-attention. A sequence classification head consisting of a pre-classifier dense layer (768→768 with ReLU activation), dropout (p=0.2), and a final classifier layer (768→2) was added on top of the encoder. All parameters were fine-tuned end-to-end.DistilBERT was selected due to its strong contextual language understanding, reduced computational cost, and suitability for real-world deployment compared to larger transformer models such as BERT.

Text was tokenized using the DistilBERT WordPiece tokenizer with a maximum sequence length of 256 tokens, padding, and truncation. Training was conducted for 3 epochs with a batch size of 16 using the AdamW optimizer with learning rate 2×10^{-5} and weight decay 0.01. A linear warmup schedule over the first 10% of training steps (660 warmup steps out of 6,606 total) was applied to stabilize early training. Gradient clipping with maximum norm 1.0 was used to prevent exploding gradients. The best model checkpoint was selected based on validation F1-score.

F. Evaluation Metrics

All models were evaluated on the held-out test set using accuracy, precision, recall, and F1-score. F1-score was used as the primary metric as it balances precision and recall, making it more informative than accuracy alone in binary classification settings. Confusion matrices were generated to analyze error patterns across all models.

G. Explainable AI

Post-hoc explanations were generated for individual DistilBERT predictions using two complementary methods. LIME [14] generates local explanations by creating 300 perturbed versions of each input article and fitting a linear surrogate model to approximate the decision boundary locally. SHAP [15] applies the Partition Explainer with Shapley values to assign mathematically grounded token-level contribution scores. Both methods were applied to one correctly predicted fake article and one correctly predicted real article from the test set to demonstrate contrasting explanation patterns.

IV. Results

A. Baseline Performance

Table I summarizes the test set performance of all three models. The TF-IDF + Logistic Regression baseline achieved an F1-score of 0.9925 with 32 total misclassifications out of 4,404 test articles. The TF-IDF + Linear SVM baseline achieved an F1-score of 0.9974 with only 11 misclassifications. Fig. 2 shows the confusion matrices for both baseline models.

TABLE I
Test Set Performance Comparison

Model	Acc	Prec	Rec	F1
TF-IDF + LR	0.9927	0.9901	0.9948	0.9925
TF-IDF + SVM	0.9975	0.9967	0.9981	0.9974
DistilBERT	**0.9995**	**1.0000**	**0.9991**	**0.9995**

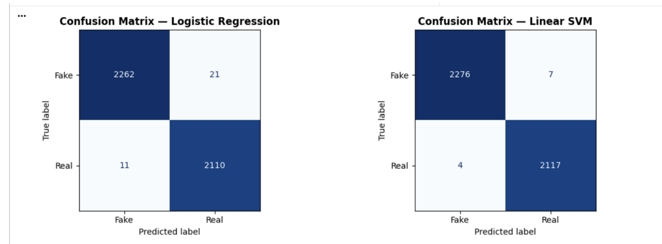

Fig. 2. Confusion matrices for TF-IDF + Logistic Regression (left) and TF-IDF + Linear SVM (right) on the test set.

B. DistilBERT Performance

The fine-tuned DistilBERT model achieved the highest performance across all metrics with an accuracy of 0.9995, perfect precision of 1.0000, recall of 0.9991, and F1-score of 0.9995. Only 2 articles were misclassified — both real articles incorrectly predicted as fake. Zero fake articles were incorrectly classified as real, resulting in perfect precision. Fig. 3 shows the DistilBERT confusion matrix and Fig. 4 shows the training curves across all 3 epochs.

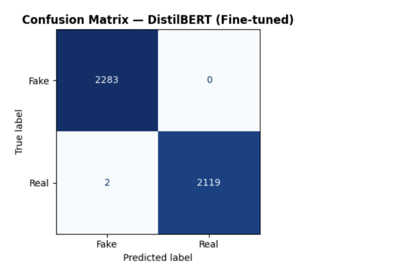

Fig. 3. Confusion matrix for fine-tuned DistilBERT showing only 2 misclassifications out of 4,404 test articles.

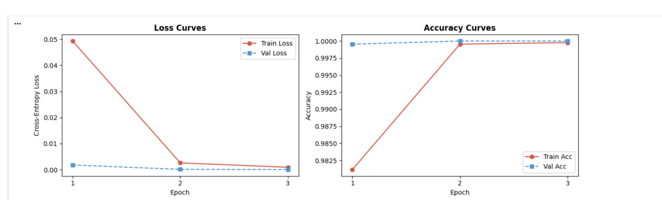

Fig. 4. Training and validation loss (left) and accuracy (right) curves across 3 epochs. The model converged rapidly with no signs of overfitting.

C. Model Comparison

Fig. 5 presents a side-by-side comparison of all three models across all four evaluation metrics. DistilBERT consistently outperforms both baselines across every metric, confirming the advantage of contextual transformer representations over surface-level TF-IDF features.

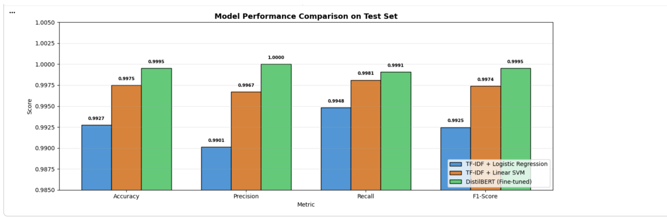

Fig. 5. Model performance comparison on the test set across all four evaluation metrics. DistilBERT achieves the highest scores across all metrics.

D. Explainable AI Results

Fig. 6 and Fig. 7 show the LIME explanations for a correctly predicted fake and real article respectively. For the fake article (predicted with 99.56% confidence), words such as "candidate," "fits," and "america" contributed negatively toward the real class. For the real article (predicted with 100% confidence), "nairobi" and "reuters" contributed most strongly toward the real class.These explanation results indicate that the model partially relies on source-specific signals (e.g., Reuters datelines) in addition to linguistic features, highlighting both the effectiveness and potential limitations of the learned representations.

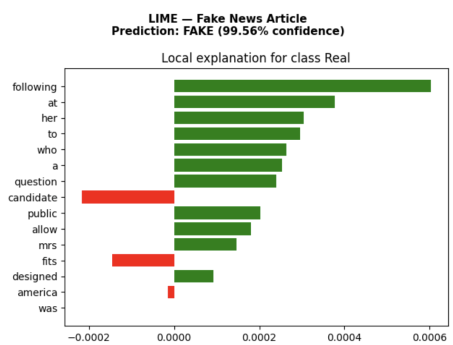

Fig. 6. LIME explanation for a fake news article (99.56% confidence). Red bars push toward Fake, green bars toward Real.

Fig. 8 shows the SHAP token-level explanations for both articles. The fake article received $f_{\text{Real}} = 0.004$ with pink highlighting throughout, while the real article received $f_{\text{Real}} = 0.999992$ with "reuters" dominating in dark blue.

V. Discussion

A. Model Performance Analysis

All three models achieved remarkably high performance, with F1-scores exceeding 0.99. This result is partly explained by the structured nature of the dataset: real articles consistently originate from Reuters wire service reports with formal datelines and geographic specificity, while fake articles exhibit more sensationalist and politically charged language. These stylistic differences create strong surface-level signals

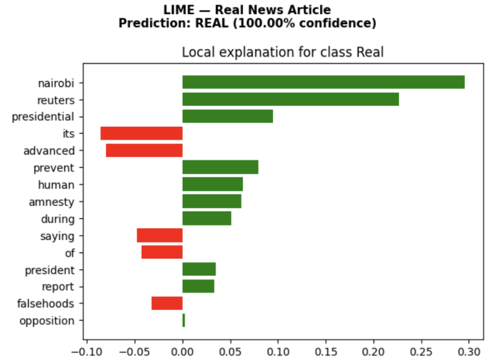

Fig. 7. LIME explanation for a real news article (100% confidence). "nairobi" and "reuters" are the strongest Real class indicators.

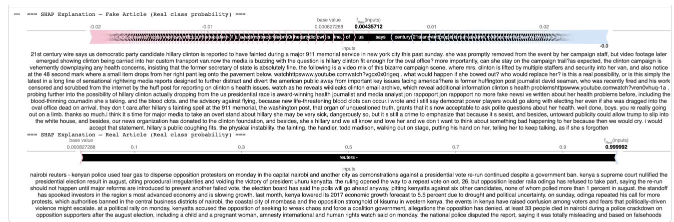

Fig. 8. SHAP explanations for fake (top) and real (bottom) articles. Pink indicates fake signals, dark blue indicates real signals.

that even TF-IDF-based models can exploit effectively.This suggests that while high accuracy is achieved, the model may not fully generalize to datasets where such source-specific patterns are absent.

Despite this, DistilBERT outperforms both baselines across all metrics, confirming that contextual transformer representations capture richer linguistic patterns beyond what TF-IDF surface features encode. The perfect precision of 1.0000 is particularly notable — every article the model predicted as real was genuinely real, with no false positives. In a misinformation detection context, false positives (flagging real news as fake) are potentially more harmful than false negatives, making this result practically significant.

B. XAI Insights

The LIME and SHAP results reveal an important behavioral insight: the model has partially learned to rely on source attribution signals (Reuters datelines) rather than purely content-level linguistic patterns. While this strategy works well within the dataset, it represents a vulnerability — a sophisticated adversary could potentially mimic wire service formatting to evade detection. Future work should investigate whether removing source attribution signals from training data forces the model to learn more robust content-level features.

C. Limitations

Several limitations of this study should be acknowledged. First, the dataset covers only 2015–2017 US political news, limiting generalization to other domains, languages, and time periods. Second, evaluation was conducted on a single dataset; cross-dataset testing on benchmarks such as the LIAR dataset [4] or FakeNewsNet would provide a more rigorous assessment of generalization. Third, hyperparameters were not exhaustively tuned due to compute constraints. Fourth, LIME explanations used 300 perturbation samples rather than the recommended 500 or more, which may reduce explanation stability. Additionally, the model may be sensitive to dataset-specific formatting patterns.

VI. Conclusion and Future Work

This paper presented a fake news detection system combining fine-tuned DistilBERT with LIME and SHAP explainability techniques. The proposed system achieved an F1-score of 0.9995 and perfect precision on a test set of 4,404 articles, outperforming TF-IDF baselines with Logistic Regression (F1: 0.9925) and Linear SVM (F1: 0.9974). XAI analysis revealed that the model associates Reuters wire service language with real news and sensationalist political language with fake news, providing actionable transparency for end users.This work demonstrates that combining high predictive performance with interpretability is essential for deploying trustworthy AI systems in misinformation detection.

Future work should address the identified limitations through: (1) cross-dataset evaluation on LIAR and FakeNewsNet benchmarks to assess generalization; (2) training without source attribution signals to encourage content-level feature learning; (3) systematic hyperparameter optimization; (4) extension to multimodal fake news detection incorporating images and social context; and (5) global XAI analysis across the full test set to characterize overall model behavior rather than individual predictions.

Acknowledgment

The authors would like to thank Dr. Benyamin Haghi and the CECS 551 course staff at California State University, Long Beach for their guidance throughout this project.

References

[1] F. Olan, U. Jayawickrama, E. O. Arakpogun, J. Suklan, and S. Liu, "Fake news on social media: the impact on society," *Information Systems Frontiers*, vol. 26, no. 2, pp. 443–458, 2024.

[2] N. Pérez-Rosas, B. Kleinberg, A. Lefevre, and R. Mihalcea, "Automatic detection of fake news," in *Proc. COLING*, 2018, pp. 3391–3401.

[3] C. Chen and K. Shu, "Can LLM-generated misinformation be detected?" in *Proc. ICLR*, 2024.

[4] W. Y. Wang, "'Liar, Liar Pants on Fire': A new benchmark dataset for fake news detection," in *Proc. ACL*, 2017, pp. 422–426.

[5] J. Alghamdi, S. Luo, and Y. Lin, "A comprehensive survey on machine learning approaches for fake news detection," *Multimedia Tools and Applications*, vol. 83, no. 17, pp. 51009–51067, 2024.

[6] H. Rashkin, E. Choi, J. Y. Jang, S. Volkova, and Y. Choi, "Truth of varying shades: Analyzing language in fake news and political fact-checking," in *Proc. EMNLP*, 2017, pp. 2931–2937.

[7] A. Mallik and S. Kumar, "Word2Vec and LSTM based deep learning technique for context-free fake news detection," *Multimedia Tools and Applications*, vol. 83, no. 1, pp. 919–940, 2024.

[8] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL*, 2019, pp. 4171–4186.

[9] R. K. Kaliyar, A. Goswami, P. Narang, and S. Sinha, "FakeBERT: Fake news detection in social media with a BERT-based deep learning approach," *Multimedia Tools and Applications*, vol. 80, pp. 11765–11788, 2021.

[10] M. Farokhian, V. Rafe, and H. Veisi, "Fake news detection using dual BERT deep neural networks," *Multimedia Tools and Applications*, vol. 83, no. 15, pp. 43831–43848, 2024.

[11] S. Raza, D. Paulen-Patterson, and C. Ding, "Fake news detection: comparative evaluation of BERT-like models and large language models with generative AI-annotated data," arXiv:2412.14276, 2024.

[12] D. Tzannetos, B. Marroquín, and P. Tsantilas, "Fake news detection and classification: a comparative study of CNNs, LLMs, and NLP models," *Future Internet*, vol. 17, no. 1, p. 28, Jan. 2025.

[13] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter," arXiv:1910.01108, 2019.

[14] M. T. Ribeiro, S. Singh, and C. Guestrin, "'Why should I trust you?': Explaining the predictions of any classifier," in *Proc. KDD*, 2016, pp. 1135–1144.

[15] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. NeurIPS*, 2017, pp. 4765–4774.

[16] E. S. Albtoush, K. H. Gan, and S. A. A. Alrababa, "New explainability method for BERT-based model in fake news detection," *PeerJ Computer Science*, vol. 11, e2693, Mar. 2025.

[17] E. Hashmi, S. Y. Yayilgan, M. M. Yamin, S. Ali, and M. Abomhara, "Advancing fake news detection: hybrid deep learning with FastText and explainable AI," *IEEE Access*, 2024.

[18] C. Bisaillon, "Fake and real news dataset," Kaggle, 2020. [Online]. Available: https://www.kaggle.com/datasets/clmentbisaillon/fake-and-real-news-dataset